

MA 403 Midsemester Exam – September 19, 2025

Total marks: 30

Instructions: You must justify all your answers unless otherwise mentioned. If using theorems/ facts proved in class, provide the complete statement or precise name (e.g., “the Bolzano-Weierstrass theorem”). If no other metric is mentioned, take $\mathbb{R}^n$ to be $\mathbb{R}^n$ with the Euclidean norm.

Answer all sub-parts of each question together.

Question 1. (6 marks) For each of the statements below, write whether it is TRUE or FALSE.

Only for this question, do not explain your answer.

- (a) There is a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(\mathbb{R}) = \mathbb{Q}$.
- (b) There is a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f((0, 1)) = (0, 1]$.
- (c) The function $d(x, y) = |2x - y|$ for $x, y \in \mathbb{R}$, defines a metric on $\mathbb{R}$.
- (d) The set $\mathbb{Z} \subset \mathbb{R}$ has no accumulation points.
- (e) If $S \subset \mathbb{R}$ is nonempty with $b = \sup S$, then for every $\varepsilon > 0$ there exists $x \in S$ satisfying $x \leq b < x + \varepsilon$.
- (f) If $\mathcal{F} = \{I_\alpha : \alpha \in A\}$ is a family of open intervals in $\mathbb{R}$ which are pairwise disjoint, i.e., $I_\alpha \cap I_\beta = \emptyset$ whenever $\alpha \neq \beta$, then A is a countable set.

Question 2.

- (a) **(3 marks)** Let S be an open set in $\mathbb{R}^n$, and let $x_0 \in \mathbb{R}^n$ be fixed. Let T be the set defined as $T := \{x_0 + y : y \in S\}$. Show that T is an open set.
- (b) **(3 marks)** If $f : \mathbb{R} \rightarrow \mathbb{R}$ is given by $f(x) = x^2$, prove that f is not uniformly continuous on $\mathbb{R}$.

Question 3.

- (a) **(2 marks)** State the Heine – Borel theorem.
- (b) **(2 marks)** If $a \in \mathbb{R}^n$ and $r > 0$, prove that the set $\overline{B}(a; r) := \{x \in \mathbb{R}^n : \|x - a\| \leq r\}$ is a closed set.
- (c) **(3 marks)** Let S be a bounded subset of $\mathbb{R}^n$. Let $\varepsilon > 0$. Prove that S can be covered by a finite number of balls of radius ε .

Question 4.

- (a) **(2 marks)** Define what it means for a sequence $\{x_n\}_{n \geq 1}$ to converge to a point p in a metric space (M, d) .
- (b) **(3 marks)** Let $x \in \mathbb{R}$. Let $\{x_n\}_{n \geq 1}$ be a sequence of real numbers such that $\{x_n\}_{n \geq 1}$ converges to x . Consider the sequence $\{s_n\}_{n \geq 1}$ of arithmetic means, defined by

$$s_n = \frac{1}{n}(x_1 + \cdots + x_n).$$

Prove that the sequence $\{s_n\}_{n \geq 1}$ also converges to x .

Question 5. Consider the metric on $\mathbb{R}^n$ given by $d(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|$. Let $\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}$ denote the Euclidean norm on $\mathbb{R}^n$. Here $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n) \in \mathbb{R}^n$.

Write $B_d(a; r)$ for an open ball in the metric space $(\mathbb{R}^n, d)$, i.e. $B_d(a; r) = \{x \in \mathbb{R}^n : d(a, x) < r\}$.

Write $B(a; r)$ for an open ball in $\mathbb{R}^n$ with the Euclidean norm, i.e. $B(a; r) = \{x \in \mathbb{R}^n : \|a - x\| < r\}$.

- (a) **(2 marks)** Prove that $d(x, y) \leq \|x - y\| \leq \sqrt{n} d(x, y)$.
- (b) **(2 marks)** Prove that for $a \in \mathbb{R}^n$ and $r > 0$, we have $B_d(a; r) \subset B(a; \sqrt{n} r)$ and $B(a; r) \subset B_d(a; r)$.
- (c) **(2 marks)** Let $S \subset \mathbb{R}^n$. Prove that S is open in $\mathbb{R}^n$ with respect to the Euclidean norm if and only if S is open in the metric space $(\mathbb{R}^n, d)$.